2015 Redesign of the Canadian Community Health Survey – Annual Component

In an ongoing effort to ensure data produced from the Canadian Community Health Survey – Annual Component remains relevant, is of the highest quality and is collected as effectively and efficiently as possible, major redesigns for the survey are planned every 6 to 8 years. In 2012, work began on the second major redesign (the first being in 2007) for the 2015 cycle of the CCHS. Part of this work included consultations held with federal, provincial and territorial share partners, health region authorities and academics. This resulted in major changes for the 2015 CCHS including changes to content, methodology and collection strategies. The first release of redesigned CCHS annual data was in March, 2017. This document will outline the changes to the 2015 CCHS, and how those changes could impact the user.

Changes to content:

Between 2007 and 2014, the common content (core and theme) in the CCHS remained relatively stable and priority was given to maintaining comparability between cycles. As part of the redesign project, the adoption of a new content plan in 2015 offered a unique opportunity to modernize and improve the CCHS content. After extensive consultations with key experts and federal, provincial and regional stakeholders, significant content changes were made to most modules (mainly the core and theme modules). Roughly 70% of all modules that existed before the redesign have changed to some extent. These changes vary in magnitude and include minor changes to concepts and vocabulary, revised response categories, and major conceptual changes.

Along with the modules themselves, variable naming conventions have also changed beginning in 2015. The naming convention used for variables in the 2015 CCHS use a maximum of eight characters. Please refer to Chapter 3 of the CCHS User Guide for a more detailed explanation.

Despite having a long-term content plan, changes to the plan are still possible. Changes can take place in the face of shifting data requirements or for operational reasons. All changes to the content plan require the approval of the Canadian Population Health Surveys Program (CPHSP) steering committee[1].

Survey frames:

Two frames are used for the CCHS starting in 2015. The samples for the youth population aged 12 to 17 are drawn from the Canada Child Tax Benefit (CCTB) sample frame, which was new for CCHS. Samples for the adult population aged 18 years and over are drawn from the same Area Frame used by the Labour Force Survey, which was also used by the CCHS for part of its sample prior to the redesign. Having distinct frames for separate populations offers improved design efficiency by reducing design effects. A reduction in design effects translates to equivalent or better quality estimates with fewer interviews.

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[1] The CPHSP steering committee is an interdepartmental working group formed with nine representatives from Statistics Canada's CCHS team, Health Canada and the Public Health Agency of Canada. These three organizations are represented on the CPHSP steering committee.

Canada Child Tax Benefit frame

The Canada Child Tax Benefit frame (CCTB) is used to sample persons aged 12 to 17 years. Youth are randomly selected from this frame to complete the survey before collection starts.The main advantage in using the CCTB as a frame is that it facilitates direct sampling of children of specific ages. As well, almost all children are eligible to receive the CCTB. The monthly coverage of children, i.e., the proportion of children receiving monthly benefits is about 79%. This increases to 93% in the case of annual coverage because, in addition to children receiving benefits, the frame also includes children whose benefit applications were denied. Thus, only families who have not submitted applications (7%) are not included on the frame. Included in that 7% are children and youth living in foster care, as they are not covered by the CCTB. Using the CCTB as a frame allows the CCHS to better target a sufficient number of youth therefore increasing the precision of youth estimates.

Area Frame

To target the population aged 18 and over, a sample of dwellings is selected from the Area Frame used by the Labour Force Survey. During collection, the members of each household living in the selected dwellings are listed, and then an eligible person is randomly chosen to complete the survey.

With the latest redesign of the Labour Force Survey, the information available from the area frame was augmented using information from the Household Survey Frame Service. This service provides address, phone number, as well as some socio-economic information on 98% of households in Canada. At the time of sampling for the first collection period of the redesign, about 91% of households had mailable addresses and about 83% of had at least one phone number attached.

The addition of the information from the Household Survey Frame Service made it possible to increase the number of dwellings with phone numbers available, reducing the need for in-person interviews.

Sample allocation:

Starting in 2015, a new multi-stage sample allocation strategy was used to give relatively fair sample distribution to the provinces and health regions (HRs). Sampling for the territories was treated differently and is explained below. For provinces, the redesign uses an allocation approach by province that is independent of the number of HRs. If changes occur in the HR boundaries for a given province, the impact will be only on the province and the HRs within that province. Also, a province with fewer HRs will not be penalized compared to another one with similar population size. This new design provides sufficient sample by province and sufficient sample size by HR in order to produce good estimates for a minimum prevalence at each level. This strategy follows the principle of having an allocation by province independent of the number of HRs and sufficient sample in order to produce good estimates for a minimum prevalence in each province and in each HR.

The following are the main outcomes of this new approach:

- Provincial sample sizes are independent of the number of HRs, it only depends on the population size. This is fairer between provinces given that the size and the number of HRs vary from one province to another.
- Changing HR boundaries within a province will not impact the province's sample allocation.
- Each province has enough sample to produce good estimates for a minimum prevalence of 2% using 2 years of data (3% for Prince Edward Island).
- Each HR has enough sample to produce good estimates for a minimum prevalence of 15% for most HRs using 2 years of data.
- The provinces are responsible for their HRs boundaries and have the flexibility to change them over time or to group them to improve the precision if needed.
- The use of a maximum sampling fraction of 1 out of 10 households annually will reduce the quality in the small HRs affected. Grouping is recommended in these areas.

This redesigned sample allocation strategy is more stable and robust as it allows for similar quality estimates for all HRs regardless of any future HR boundary changes that may occur within a given province.

Sample allocation for the three territories is done in a different way. Each territory corresponds to a single HR and the sample is allocated to the communities that compose the territory in a way that ensures the sample is representative of each territory over a two-year period. This means that one-year theme content will not be available for the territories on the two year file unless they have selected the same modules as optional content for the other corresponding year. Users wanting truly national representative estimates will have to use the two-year file, in order to have the territories included.

Collection:

In 2015, a dynamic telephone number strategy was developed to maximize telephone interviews to take advantage of the increased availability of telephone numbers, including cell phone numbers, linked to the frames. This change in collection strategy meant that for the 18+ portion of the sample, telephone interviews are prioritised over personal interviews when possible. For the 12-17 portion of the sample, all interviews are conducted over the phone.

For the sample aged 18+, personal interviews may be used as a non-response follow-up as required and when feasible from a cost and resource perspective. Personal Interviews may also be used to contact households where a telephone number could not be found. This new sequential multi-mode approach and the improved list of associated telephone numbers will lead to a reduction in the need and use of personal interviews, which will translate into cost savings, as telephone interviews are less costly.

Weighting and estimation:

In 2015, due to the change in frame used to select the 12-17 year olds, household weights are produced only for the sample of respondents aged 18+ selected from the area frame, who were sub-selected from a representative sample of Canadian households. These households also contain 12-17 year olds, so no coverage of Canadian households has been lost. Household

level variables such as food security indicators and household incomes can be used as correlates to health indicators for the sample of 12-17 year old respondents.

Bootstrap weight files now contain 1000 replicates. The additional replicates will provide an increase in precision of the estimates of variance in some cases.

Quality indicators:

As with previous years, the CCHS recommends that statistical analysis be performed on weighted estimates and make use of the appropriate quality indicator. Beginning in 2015, the recommended quality threshold for coefficients of variation (CV) has changed from 33.3% to 35.0%. That is, if an estimate has a CV above 35.0%, it must be marked as unfit for release according to Statistics Canada. The change to the CV threshold was made to adhere to a new Statistics Canada common tool for estimation. For the complete list of CV thresholds and quality indicators, see Table 10.1 of the CCHS User Guide.

Target Population:

Since its inception in 2001, the CCHS has had a target population of respondents aged 12 and over (among other criteria). One of the major objectives of the 2015 CCHS Redesign was to explore options for collecting health data for children under 12. To address this, a feasibility study reviewed several options, including collecting this information using the CCHS itself, creating a CCHS follow up survey or developing a standalone survey. The study recommended that a standalone survey on children's health would be the best option. This new survey, called the Canadian Health Survey on Children and Youth (CHSCY), conducted a pilot test in 2016, and its first full collection cycle is planned for 2018-2019. Please refer to the following link for more information on the CHSCY.

http://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=5233

Impacts on the User - Comparability between past cycles and the 2015 redesign:

A major redesign is an opportunity to make significant changes to improve a survey. Changes to the 2015 CCHS include the use of higher quality frames, the adoption of more efficient collection and sample allocation strategies and updated and modernised content. These changes will help the survey produce higher quality data while making the survey more efficient, however these changes impact the analytical use of previous CCHS data sets.

As a result of the changes from the 2015 CCHS redesign, combining cycles of CCHS data from before and after the redesign (e.g., combining 2014 and 2015 annual files) is not recommended, and caution should be taken when comparing estimates across those years. Even estimates derived from content that has remained unchanged, are subject to the potential impacts of the other major changes to the survey (i.e. new survey frames and collection methods) and may not necessarily be comparable with past cycles. It would be very difficult to ascertain whether any changes or consistencies between estimates pre and post redesign are a reflection of the true population characteristic being examined, or the effect of the significant methodological and operational changes made to the survey.